

Tristan Louis

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EDUCATION

Queens College

Bachelor of Arts in Computer Science

Expected to Graduate: December 2026

EXPERIENCE

Data Science Intern

Dakdan Worldwide—Remote

August 2025 - Dec 2025

- Engineered automated Python and Pandas ETL pipelines to ingest and transform marketing data, accelerating data-driven sponsorship and promotion scheduling.
- Automated data validation scripts across 15+ recurring workflows, eliminating manual data entry errors to support revenue and marketing reporting.
- Designed and tuned PostgreSQL database models and query execution paths, improving analytical dashboard loading speed by 25%.

Software Development Intern

OpenQuantify—Remote

June 2025 - Dec 2025

- Developed 8+ REST API endpoints using FastAPI and Flask to deliver high-throughput data access for AI-driven applications.
- Re-architected the backend service into 3 decoupled tiers (API, logic, and data), reducing code duplication across 3 core modules.
- Applied targeted B-Tree indexes on PostgreSQL read paths, cutting query latency by 30% under active workload.
- Built backend data pipelines connecting microservices to CesiumJS, streaming 3D spatial telemetry for real-time digital twin simulations.

Data Engineer Fellow

Open Avenues Foundation: The Build Fellowship Highlight Reel: Automated Sports Content Generation

September 2024 – December 2024

- Built an automated video processing pipeline with OpenCV and XGBoost to generate sports highlights.
- Developed computer vision feature extraction to track ball movement and identify key game events in real time.
- Boosted highlight detection accuracy from 67% to 89% using targeted feature engineering and model tuning.

PROJECTS

PathBot Inventory Assistant

Python, FastAPI, PostgreSQL, SQLAlchemy, BeautifulSoup

- Migrated data storage from JSON files to PostgreSQL, boosting search speeds by ~60%.
- Built a BeautifulSoup web-scraping pipeline to parse and upsert 500+ inventory items daily.
- Added multi-modal query parsing to handle unstructured search inputs, reducing lookup errors by 35%.

Real-Time Global News Intelligence Pipeline & Analytics Dashboard

PySpark, PostgreSQL, FastAPI, Streamlit

- Built a PySpark ETL pipeline processing 1,000+ daily news articles into PostgreSQL.
- Designed relational schemas and ranking algorithms to compute Top-10 summaries in under 2 seconds.
- Connected FastAPI endpoints to a Streamlit dashboard with real-time topic filtering.

TECHNICAL SKILLS

- Languages:** Python, SQL
- Data Engineering & Databases:** PySpark, PostgreSQL, SQL, Data Modeling, Schema Design, ETL/ELT Pipelines, Pandas, NumPy, SQLAlchemy
- Backend & Cloud Tools:** FastAPI, REST APIs, Git, Azure, AWS (S3, Lambda)

CERTIFICATES

- Databricks Data Engineer Associate—Expected October 2026